

TASK 4

Customer Churn & Retention Analysis

Analyze why customers leave, identify high-risk groups, and convert findings into practical retention recommendations.

Focus: Data → Analysis → Statistical Evidence → Business Insight → Recommendation

1. Dataset & Setup

Dataset: Telco Customer Churn (IBM sample / BlastChar version). [Kaggle dataset](#)

- Use Python (Pandas, NumPy, Matplotlib) and Power BI.
- Check shape, data types, missing values, duplicates, and descriptive statistics.
- Correct obvious data-type or missing-value issues and briefly document what you changed.

Resource: [Pandas documentation](#) — useful topics: read_csv, info, describe, value_counts, groupby.

2. Churn Analysis

- Overall churn: total customers, churned customers, retained customers, and churn rate.
- Contract: compare Month-to-month, One-year, and Two-year contracts; identify the highest churn rate.
- Tenure: create 0–12, 13–24, 25–48, and 49+ month groups; identify the highest-risk group.
- Monthly charges: compare churned vs retained customers and report the average for both.
- Payment method: calculate churn rate by payment method.
- Services: choose any two (e.g., Internet Service, Tech Support, Online Security, Device Protection) and analyze churn differences.

Requirement: Use only useful charts and add a 1–2 sentence interpretation for each major result.

3. Statistical Analysis

Test 1 — Contract vs Churn

Use a Chi-Square Test of Independence.

H0: Contract and churn are independent.

H1: Contract and churn are associated.

Report: statistic, p-value, decision, and short interpretation.

[SciPy: chi2 contingency](#)

Test 2 — Monthly Charges vs Churn

Use an Independent Samples T-Test.

Compare monthly charges for churned and retained customers.

Report: group averages, t-statistic, p-value, decision, and short interpretation.

[SciPy: ttest ind](#)

Use significance level $\alpha = 0.05$.

4. Customer Segmentation & Revenue at Risk

- Create 3–4 simple rule-based segments using tenure, monthly charges, and churn. Suggested labels: New, Loyal, High-Value, High-Risk.
- For each segment, report customer count, average tenure, average monthly charges, and churn rate.
- Identify which segment needs the most retention attention and explain why.
- Calculate Monthly Revenue at Risk = sum of MonthlyCharges for churned customers.

5. Power BI Dashboard

- Create one clean dashboard page.
- KPI cards: Total Customers, Churned Customers, Churn Rate, Average Monthly Charges, Revenue at Risk.

- Create 4–5 visuals: churn by Contract, Tenure Group, Payment Method, Customer Segment, and Monthly Charges comparison.
- Add 2–3 slicers if useful (e.g., Contract, Internet Service, Payment Method).

Resources: [Microsoft Learn - Power BI](#) | [Card visual](#) | [Slicers](#)

6. Findings & Recommendations

- Write 5 key findings supported by calculated results.
- Write 5 practical recommendations, each directly linked to a finding.
- Avoid generic recommendations that are not supported by the dataset.

7. Final Deliverables

Python Notebook

Task4_Customer_Churn_Analysis.ipynb

Data preparation, churn analysis, tests, segmentation, revenue-at-risk, charts.

Power BI Dashboard

Task4_Churn_Dashboard.pbix

One dashboard page with KPI cards, 4–5 useful visuals, and optional slicers.

Short Summary

Task4_Final_Summary.pdf

Maximum 1–2 pages: findings, statistical results, important segment, revenue at risk, recommendations.

Important Instructions

- Do not build a machine-learning prediction model for this task.
- Do not copy complete Kaggle notebooks.
- Keep the analysis focused; avoid unnecessary charts.
- Interpret important results briefly and clearly.
- Recommendations must be supported by your analysis.

EXPECTED OUTCOME

Demonstrate the ability to move from raw customer data to evidence-based business decisions and retention recommendations.